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Client: Simmond's Precision

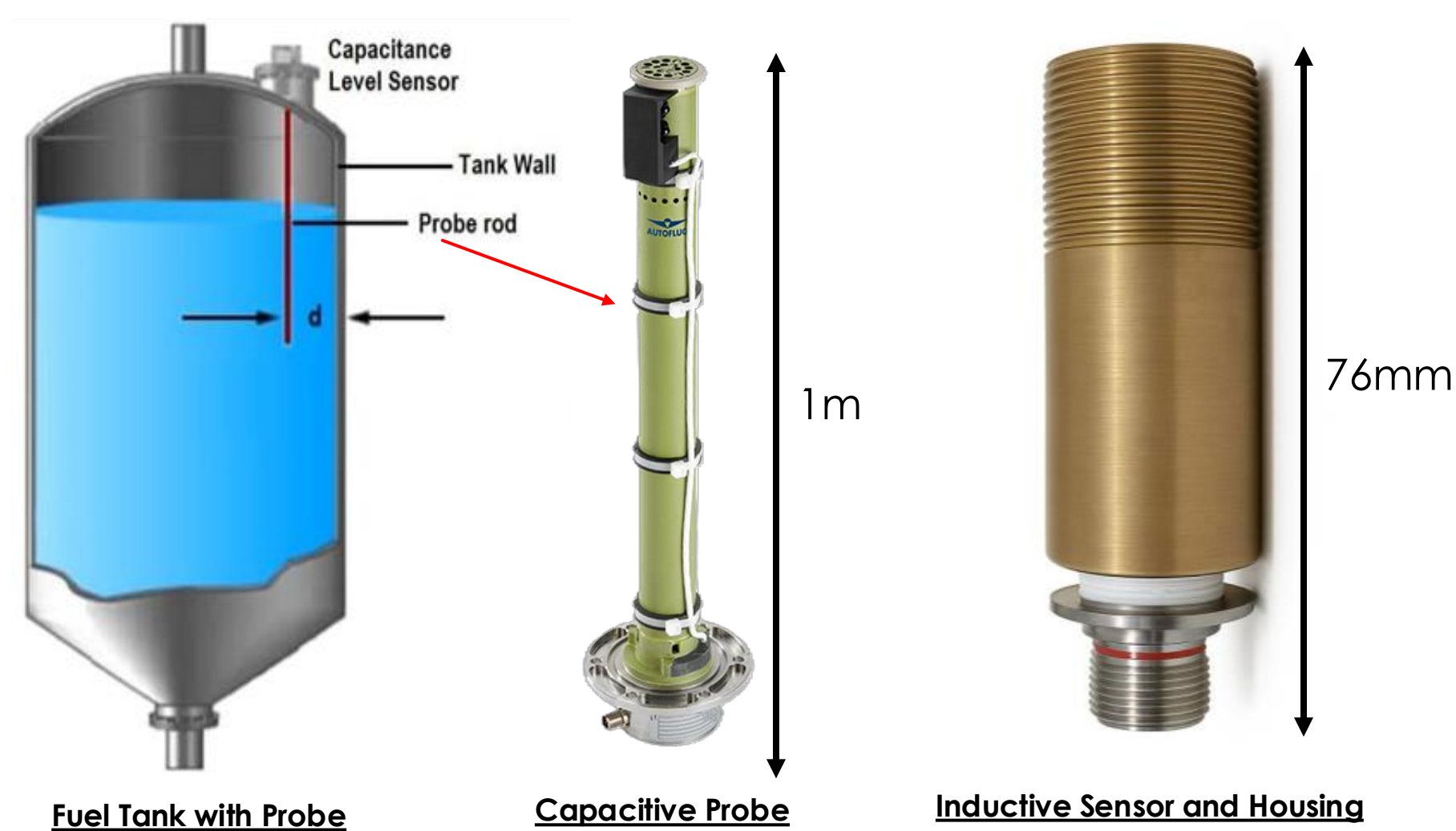
Global leader in sensor design for aerospace and defense located in Vergennes, VT.

We have been working closely with our client over the last two semesters, meeting weekly.

Define the Problem

Current standard fuel sensors are capacitive which are both bulky and mounted internal to the fuel tank. The new inductive design is both smaller and mounted externally to the fuel tank.

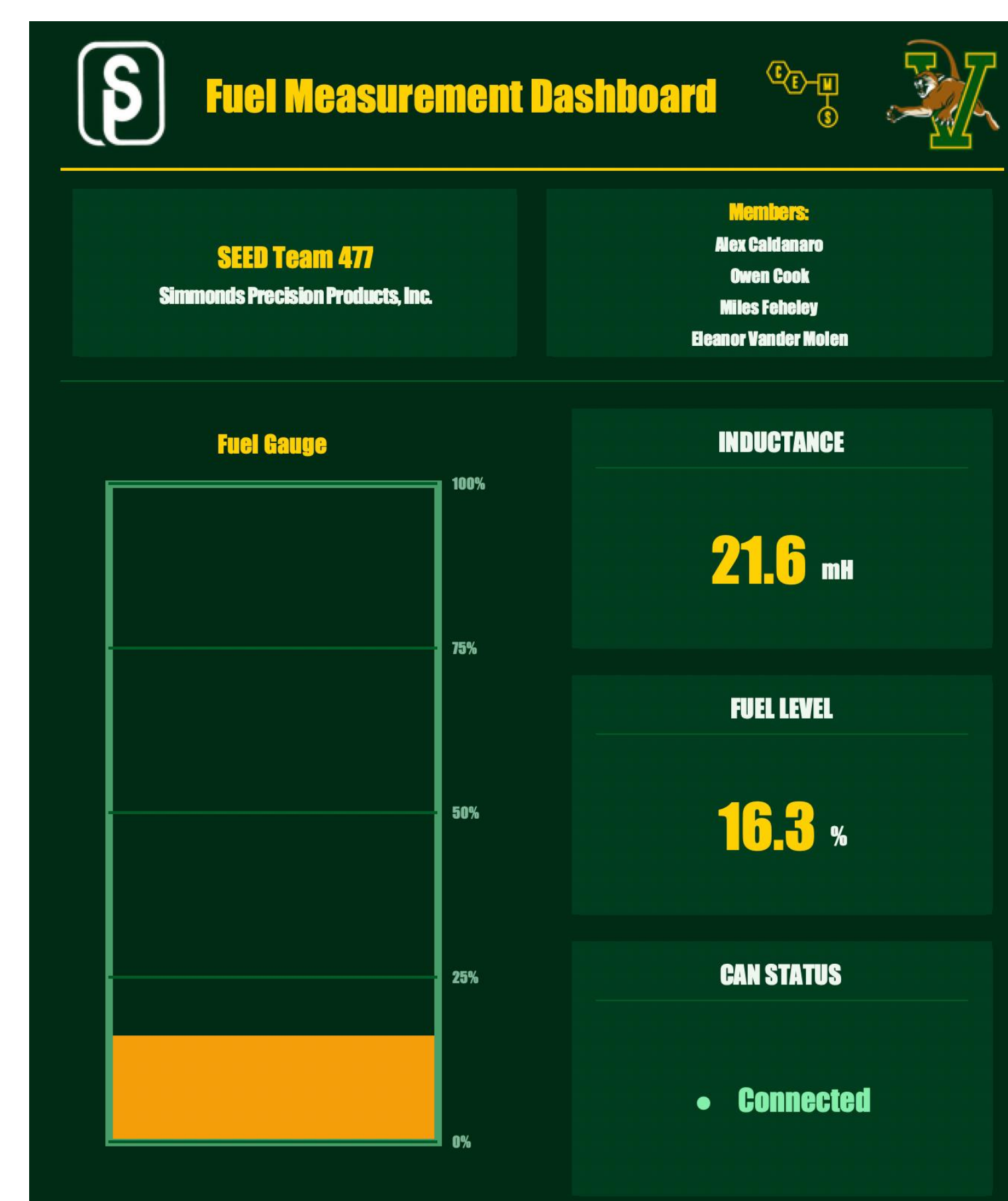
Previous year's circuitry is large and is used for more than just fuel reading. Our task is to create electronics and packaging for just the fuel reading.



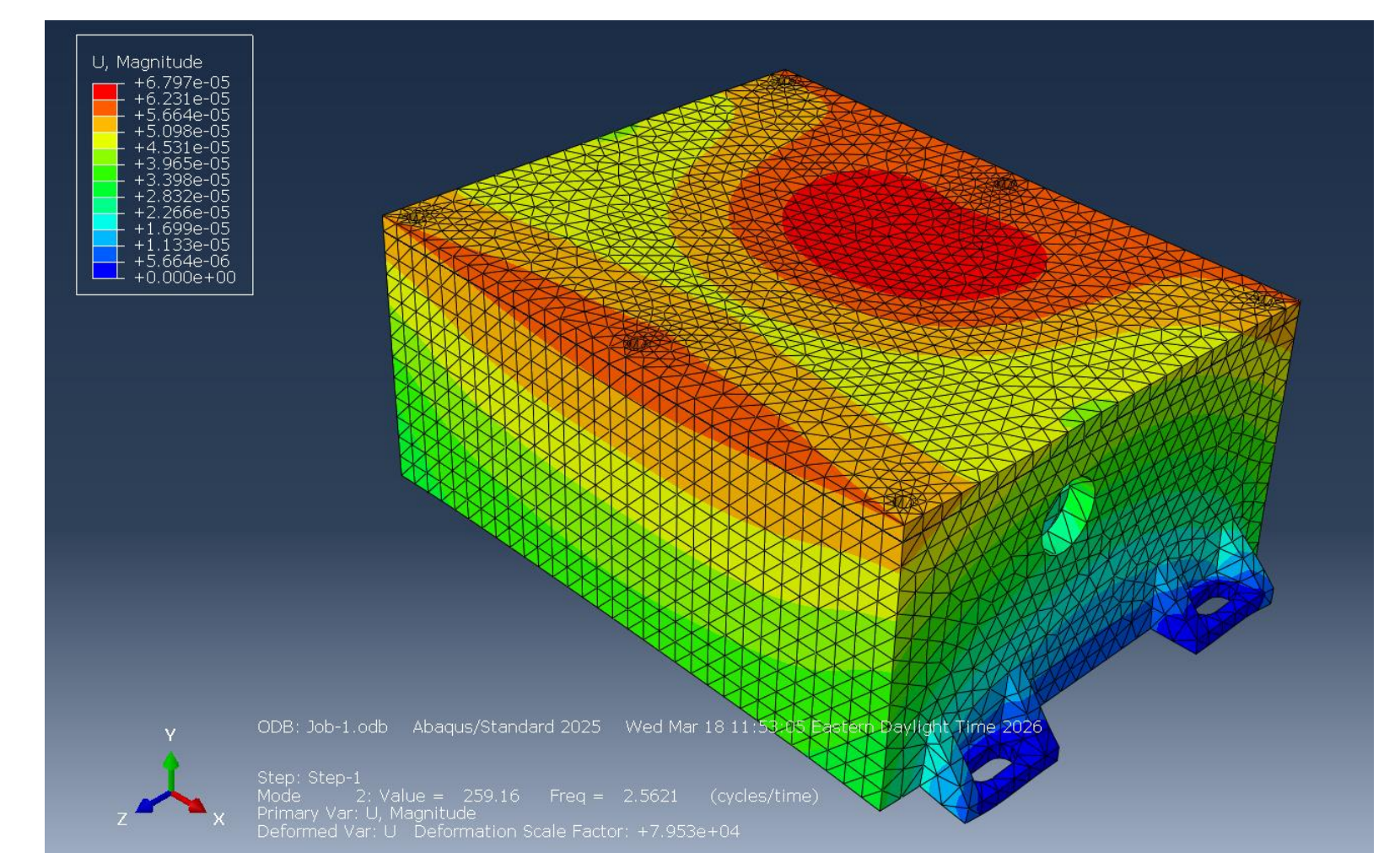
Client Requirements

- Send data over CAN protocol
- Map inductance to pressure
- Electronic components, excluding the sensor shall fit into dimensions of 900cm³
- System shall be capable of operation between 19°C and 27°C
- UI to display inductance reading read from CAN bus

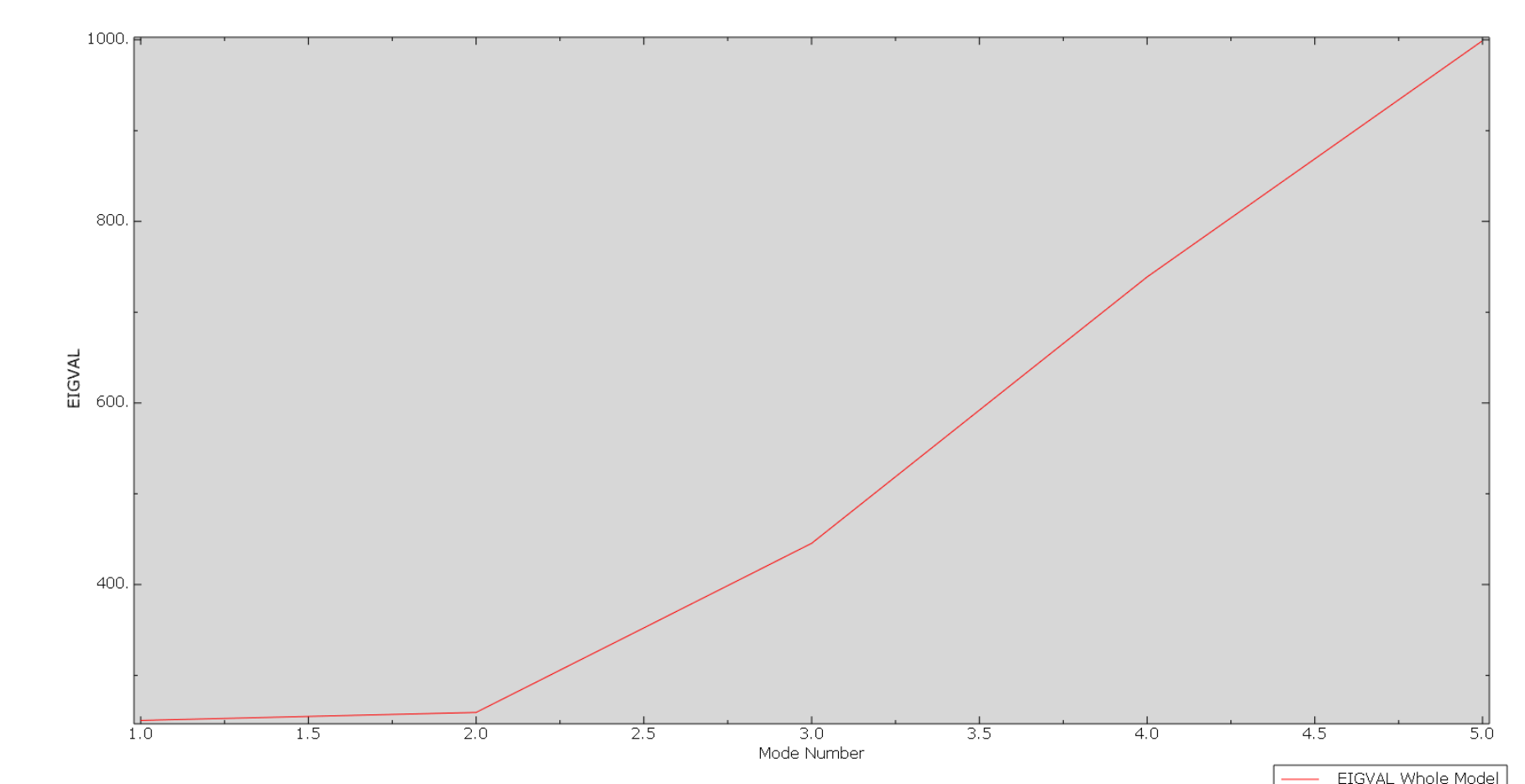
Working Design Concept



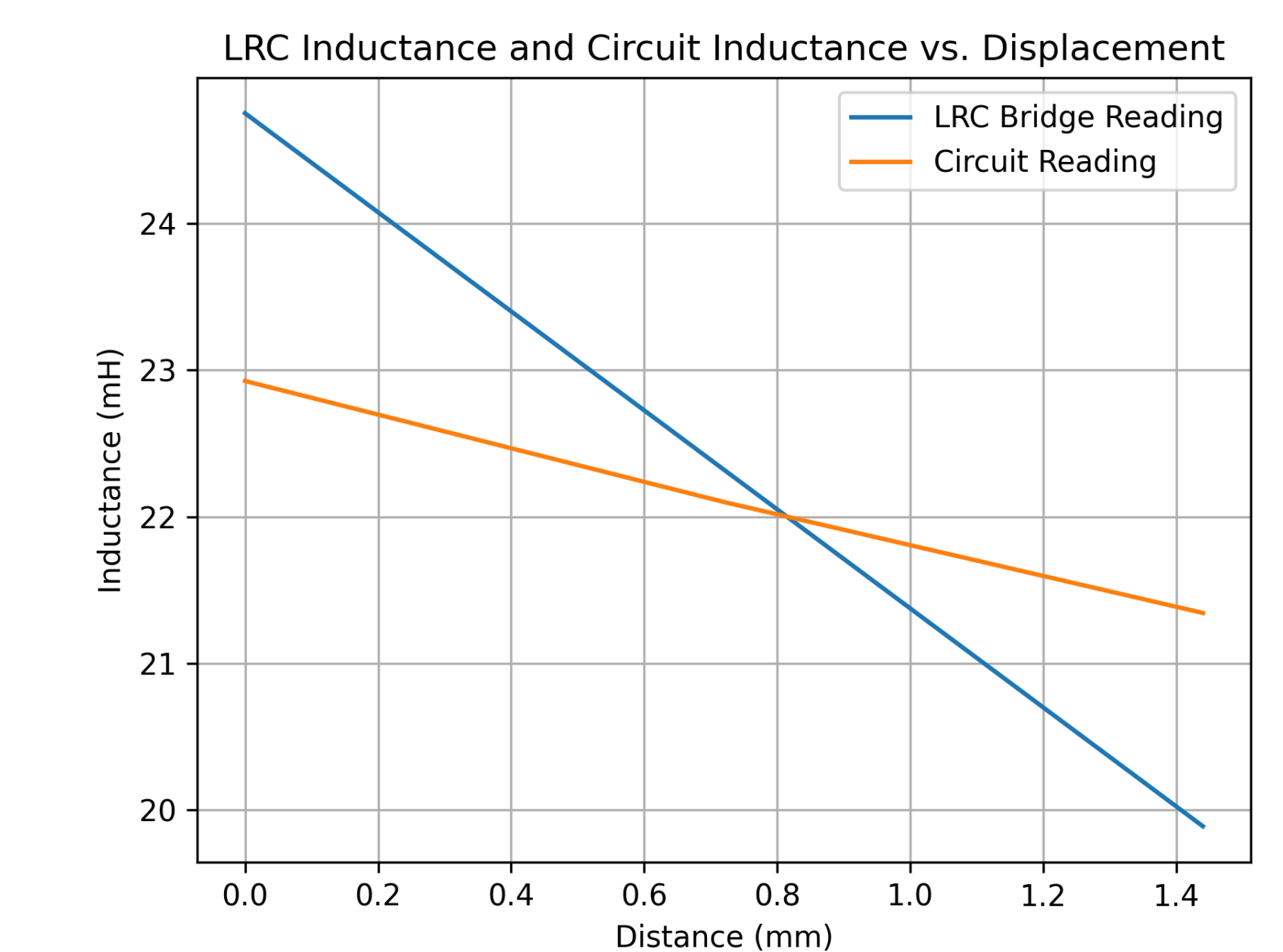
Results



Vibrations Analysis



Eigenfrequencies vs. Mode Number



Max % Error: 7.96%

Acknowledgments

- Emay Pandarakutty
- Josh Bouffard
- Peter Conklin
- Philip Buchwald
- Dr. Cosoroaba
- Keith Doyle and Keith Epstein

Future Work

- Use different IC to send the excitation signal
- Remove first stage of signal conditioner
- Have both sides of PCB assembled
- IP testing/waterproofing
- Use faster MCU to sample faster
- Use DFT in software instead of FFT
- Improve software efficiency

Conclusion

The team spent the year implementing the electronics and creating packaging for this inductive sensor. The finished prototype is able to read inductance with a maximum of 8% error and the footprint of the device is drastically smaller.